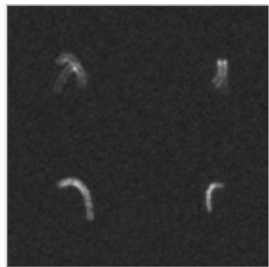


ENCODE the image is reduced once

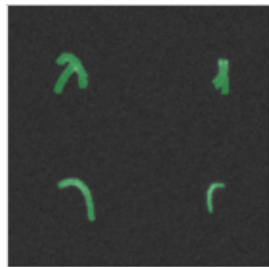
Input image



sparse fluorescence
65,536 bytes



Foreground



learned cDice U-Net
mask IoU 0.90



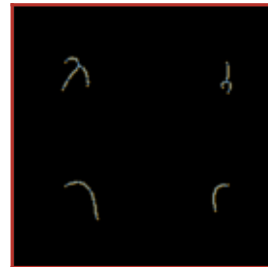
Medial axis



skeleton + distance
transform gives width



Attributed graph



144 nodes, 141 edges,
4 components



THE STORED OBJECT — the graph itself, 2,077 bytes (31× smaller than the array)

per node

x, y	w	I	θ
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position

width

intensity

orientation

× 144 nodes, delta-coded along the graph

per edge

i → j

connectivity

plus

16×16	DCT
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background field

foreground residual

no pixel array is kept; the decoder rebuilds the graph from this stream

READ THE STRUCTURE directly, no pixels touched

$$\beta_0 = 4$$

connected components, counted on the graph

$$\beta_1 = |E| - |V| + \beta_0$$

cycle rank: path redundancy of the network

mean width, edge length

per-edge morphometry in stored units

junction degrees

branching pattern of the network

*exact properties of the stored graph: no re-segmentation,
no re-tracing, identical on every read*

RENDER THE IMAGE on demand



elliptical PSFs placed at the nodes,
each aligned to its stored orientation

PSNR 30.9 dB · FG-PSNR 28.8 dB

pixel codec: bytes → decode → segment → trace → estimate descriptors (re-derived, tool-dependent)

Nanograph: bytes → read descriptors · bytes → render image (stored once, then read)